

Digital Wallet Transfer Limits & Account Tiers

Risk-Based Test Plan & Spec Analysis (Part 1)

Document: Deliverable 1 — Risk-Based Test Plan

Target Feature: Core Transfer Limits & Tiers

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1. Scope & Objective

This test plan covers the wallet transfer limits and account-tier feature as described in the seven-rule specification provided. The goal at this stage is not to enumerate every possible test case, but to surface the ambiguities that must be resolved before test design can be trusted, then define a prioritized, risk-based approach to coverage that avoids a combinatorial explosion of near-duplicate cases.

Because this feature moves money, the guiding principle throughout is: **correctness of limit enforcement and exemption logic matters more than breadth of UI or cosmetic coverage**. A missed boundary condition here is a financial-loss or fraud incident, not a cosmetic bug.

2. Open Questions for the Product Team

These would be raised before writing a single test case. Each represents a genuine ambiguity, contradiction, or unhandled edge case in the spec as written — not a nitpick. Where the spec is silent, the default assumption used to prevent blocking testing is noted.

Q1. Which timezone does “midnight” refer to?

Rule 2 says the daily limit resets at midnight but doesn't say whose midnight: user device, KYC address, provider operating timezone (e.g. WAT), or UTC. Device-timezone implementations allow users to double reset by crossing boundaries.

Default assumption: single fixed provider timezone.

Q2. How is a pending transfer counted across a midnight reset?

Rule 3 says pending transfers count toward the daily limit, but if created at 23:58 and pending at 00:02, does it consume day N permanently, get recount against day N+1 (double-counted), or get dropped? Option (b) causes under-spending issues; option (c) opens limit bypass exploits.

Q3. Is there a documented concurrency/locking strategy for remaining-limit checks?

Rule 7 describes a check against "remaining limit" as a read, but simultaneous transfers from two sessions could read the same remaining limit and both pass before either deducts. Must confirm if checks are atomic DB-level operations.

Q4. Does a reversal restore limit to the day it was created on, or to the current day?

Rule 3 returns reversed amounts to available limit. If sent Monday and reversed Wednesday, crediting Wednesday gives unintended headroom on Wednesday. Repeated send-and-reverse cycles could artificially inflate daily limits.

Q5. What happens when a scheduled transfer's 8 AM execution meets an already-exhausted limit?

Rule 5 executes transfers at 8 AM. If limit is exhausted by other transfers prior to 8 AM, does it fail silently, retry, or alert? Also, is the amount reserved at scheduling time or execution time?

Q6. How is “own linked bank account” verified, and can that verification be gamed?

Rule 4 exempts own bank transfers. This is the highest-value fraud target since it has no ceiling. Must clarify verification mechanism (name matching, micro-deposits, bank API).

Q7. Which tier limit applies to a transfer that's already pending when the user's tier changes?

Rule 6 upgrades immediately. Is a pending Tier 1 transfer grandfathered or re-evaluated under Tier 2? How are KYC downgrades handled?

Q8. Does “no per-transaction limit” for Tier 3 override the daily limit for a single transfer?

Rule 1 states Tier 3 has no per-transaction limit but a 5,000,000 daily limit. A single 6,000,000 transfer should be caught by Rule 7, but needs explicit confirmation.

Q9. Are “failed” and “reversed” transfers the same event for limit-restoration purposes?

Failed transfers never settled; reversed transfers completed and were undone. Must ensure single-source-of-truth writing so amounts are never credited twice.

Q10. What currency do the limit figures apply to, and how is a multi-currency wallet handled?

Limits lack currency units. Clarify if enforced per-currency or converted at live rates to a base currency.

3. Prioritization: What to Test First, and Why

Testing proceeds in four waves ordered by financial risk and blast radius:

- **Wave 1 — Core limit enforcement (highest risk):** Boundaries for all 3 tiers & Rule 7 rejection messages. Fully automated.
- **Wave 2 — Exemption & reset logic:** Own-linked bank exemption (Rule 4) & midnight reset boundary (Rule 2). Clock mocking used for reset testing.
- **Wave 3 — State-transition logic:** Failed/reversed restorations (Rule 3) & in-flight tier upgrades (Rule 6).
- **Wave 4 — Scheduling & concurrency (highest uncertainty):** 8 AM execution (Rule 5) & concurrent race conditions.

4. Coverage Approach: Tier × Transfer Type × Limit-State Matrix

A naive full cross-product results in 60 combinations (3 tiers × 4 types × 5 limit-states). This plan collapses the matrix down to 20–25 meaningful cells using two key techniques:

1. **Isolate the exemption path:** “Own linked bank account” is exempt across all tiers, reducing 15 cross-product cells into 3 representative tests.

2. **Orthogonal transfer types:** Wallet-to-wallet, wallet-to-other-bank, and scheduled transfers share the same limit engine. Boundary maths are thoroughly tested on core tiers and spot-checked for other types.

5. Known Gaps at This Stage

Test cases dependent on product answers to Q1–Q10 are kept as backlog placeholders to prevent encoding unverified assumptions into passing test suites.